

Your Text: 3.500 - 3.900 characters with spaces

A Vision for Textile Reinforced Concrete Structural Sections

Barzin Mobasher

The life cycle costs of structural systems is a function of raw materials, labor, manufacturing method, serviceability requirements, environmental impact, and durability. The overarching objective is to consider Textile reinforced concrete (TRC) composites as emerging lightweight construction materials with strength and ductility that may compete and outperform light gage steel and wood products. Such a goal should encompass a design that is amenable to continuous production and formability for distinct sections, so that the end product can be optimized for long-term durability, ductility, high strength and thus sustainable. These considerations permit us to make Textile Reinforced Concrete (TRC) a new cost-competitive construction material by considering four key design that include: novel manufacturing, alternative reinforcing systems, standardized sections, and design tools for various loading conditions. It is therefore imperative that new benchmarks be developed so each one of these areas are adequately addressed while their interaction may lead to composites with high strength, ductility, and stiffness.

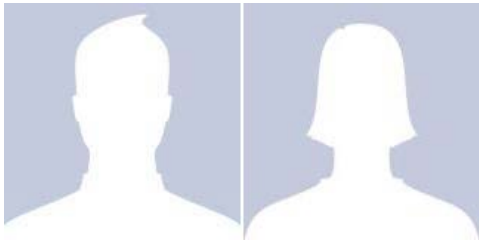
Economic incentives favor the use of polypropylene (PP) based textiles for a variety of applications in civil infrastructure systems. Textiles with surface modified hydrophilic monofilament continuous yarns are selected since their surface texture results in enhanced fiber matrix bond. The enhanced bond activates formation of parallel microcracks and results in strengthening and toughening mechanisms under tensile loading. An effective manufacturing technique was developed using automated pultrusion process for efficient production of TRC structural sections. Using a computer controlled system, continuous cement based-PP fiber composites in a variety of shapes that include W section, angles, and C-sections are manufactured.

The primary modes of mechanical response are in terms of tensile, shear, bearing, and compression buckling. Samples in tension are characterized using first cracking strength, post crack stiffness, ultimate strength and strain capacity. A range of volume fractions of 1-4% by volume of fibers are normally used resulting in tensile first cracking in the range of 2-7 MPa, ultimate strength of up to 25 MPa and strain capacity of up to 6%. The tensile hardening behavior is attributed to the fiber bridging effect which stabilizes crack growth and opening at the expense of formation of multiple, parallel, fine-cracks. This cracking network gives rise to high energy absorption both under quasi-static and dynamic loading conditions. The post crack stiffness and the corresponding damage distribution may form with a variety of fiber systems and is governed by the fiber's ability to provide a sufficient degree of bond strength. The superior mechanical properties can be utilized in structures subjected to static and dynamic loads, reinforcement laminates, or strengthening/repair materials.

The final component of the design space is in the development of generalized design tools using analytical closed form solutions for serviceability based design of tensile, compression, shear, connection, and flexure of 1-D and 2-D elements. Using a sectional analysis, parametric models for obtaining closed-form moment-curvature relationships for a cross section under various tensile or compression failure criteria can be developed. Next, a trilinear normalized moment-curvature relationship is used to obtain closed form load-Deflection relationships for a variety of beam loading cases as well as yield line collapse mechanisms. The models is used to address the flexural design, using parameters such as maximum allowable curvature, crack width, deflection, or any other serviceability based criteria based on tension softening or hardening response.

Short vita

500 - 600 characters with spaces



Your photo

354 x 472 pixel, high-resolution
image data of at least 300 dpi

Barzin Mobasher obtained his PhD. in Civil Engineering from Northwestern University in 1990 and joined Arizona State University in 1991 as an Assistant Professor. He is currently a professor after leading research programs addressing design, analysis, materials and full-scale structural testing of structural and aerospace materials. His publications include two books on fiber and textile reinforced concrete, three edited books, and more than 150 research papers in leading professional journals. His contributions in the fields of concrete materials, mechanics of toughening and durability.

Titel: max. 90 characters with spaces

A Vision for Textile Reinforced Concrete S

Cover picture:

Visible range 213 x 249 mm,
Image size 2.516 x 2.941 pixel
high-resolution image data of at least
300 dpi, possible file formats are:
JPG, PSD, TIF, EPS, PDF

Please send us only a portrait format!

Don't forget the copyright

Description + name of the
photographer/author

